



Probability and Statistics Honors

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–12 · FULL YEAR · HONORS

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in lesson order, activities, and examples from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Probability and Statistics Honors develops a scholar's ability to reason rigorously about data and evidence — the mathematics of everyday decisions. Semester 1 is about describing and modeling data: choosing the right display, measuring center and variability, and fitting linear and exponential models (and knowing why correlation is not causation). Semester 2 turns to how trustworthy data is made and used: sampling, study design, margin of error, and probability through the binomial, geometric, and normal distributions. VR and AI enhancements support deeper exploration of datasets, and honors scholars take on independent and group research projects. Throughout, the course builds precision in calculation and care in drawing conclusions — learning to recognize where data illuminates and where it misleads. Prerequisite: Algebra 2.

THE YEAR AT A GLANCE

1 Representing and Interpreting Data (Units 1–3, Sem 1)

Choosing the right way to display data; measures of center and variability; two-way frequency tables and segmented bar graphs in real-world contexts.

3 Sampling and Study Design (Units 7–8, Sem 2)

Population versus sample, why random sampling works, and how surveys, experiments, and observational studies differ — and when each is right.

5 Probability and Distributions (Units 12–14, Sem 2)

Sample spaces, permutations and combinations, conditional probability, random variables, and the binomial, geometric, and normal distributions.

2 Modeling Relationships (Units 4–6, Sem 1)

Correlation and linear models with residual analysis, the crucial difference between correlation and causation, and exponential models.

4 Statistical Inference (Units 9–11, Sem 2)

Estimating population measures, margin of error and confidence, fair decisions with probability, and evaluating data claims in the media.

The Honors difference. Greater depth and pace throughout, with independent and group research projects in which scholars collect, analyze, and present real data — and defend their conclusions.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- Quizzes and tests close each unit; assignments open at the start of each quarter, with due dates that pace the work toward quarter close.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A scientific calculator — correlation, distributions, and inference work lean on it.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, when offered live; for 2026–27 Probability and Statistics Honors is listed as an On-Demand course.